

ainer

||

om Fields

Table of Contents

1.	System Overview	3
2.	User Roles & Permissions	4
3.	Container Management	5
4.	Cost Recording — Five Sections	7
5.	Custom Sections & Extra Fields	9
6.	Client Management	10
7.	Client Wallet & Financial Tracking	11
8.	Invoicing & Accounts Receivable	13
9.	Analytics & Reports	15
10.	Pipeline / Kanban Board	17
11.	Notifications & Alerts	18
12.	Approvals Workflow	19
13.	Task Management	20
14.	Container Upload (Bulk & Single)	21
15.	Public Container Tracking	22
16.	Settings & Configuration	23
17.	WhatsApp Notifications	24
18.	Print & Export	25
19.	Security & Data Integrity	26



System Overview

The Cost Analysis Per Container system is a web-based platform purpose-built for Nigerian bonded terminal operators who manage container clearing on behalf of importers and corporate clients. Every figure is denominated in Nigerian Naira (₦). The system provides a complete financial picture — from the moment a container arrives at the terminal through to delivery and invoicing.

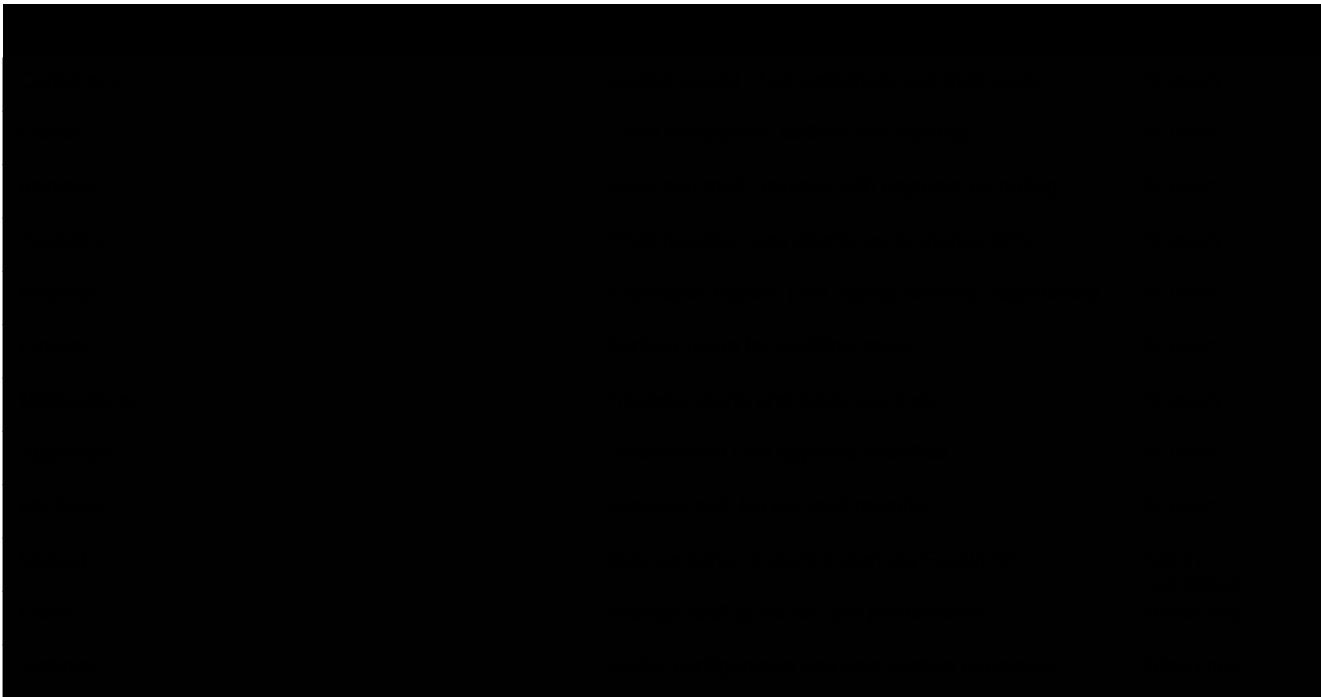
Core Design Principles

- All monetary values are displayed and stored in Nigerian Naira (₦)
- Role-based access control — admins have full access, staff have configurable permissions
- Real-time alerts for financial anomalies (negative profit, low margins, delays)
- Audit-ready — every change is logged with timestamps and user attribution
- Printable outputs for every major document (container summary, invoice, client statement, reports)
- Bulk data entry via Excel/CSV upload for both containers and clients

Technology Stack

Frontend:	React + TypeScript, Vite, Tailwind CSS, Shadcn/UI components
Backend:	Node.js (Express), TypeScript, Drizzle ORM
Database:	PostgreSQL (Replit-managed)
Authentication:	Session-based login with bcrypt password hashing
Notifications:	Twilio WhatsApp API + in-app real-time alerts
Export:	Browser-native print to PDF for all print views

System Modules at a Glance





Administrator

- ## Staff

- ## Permission Matrix

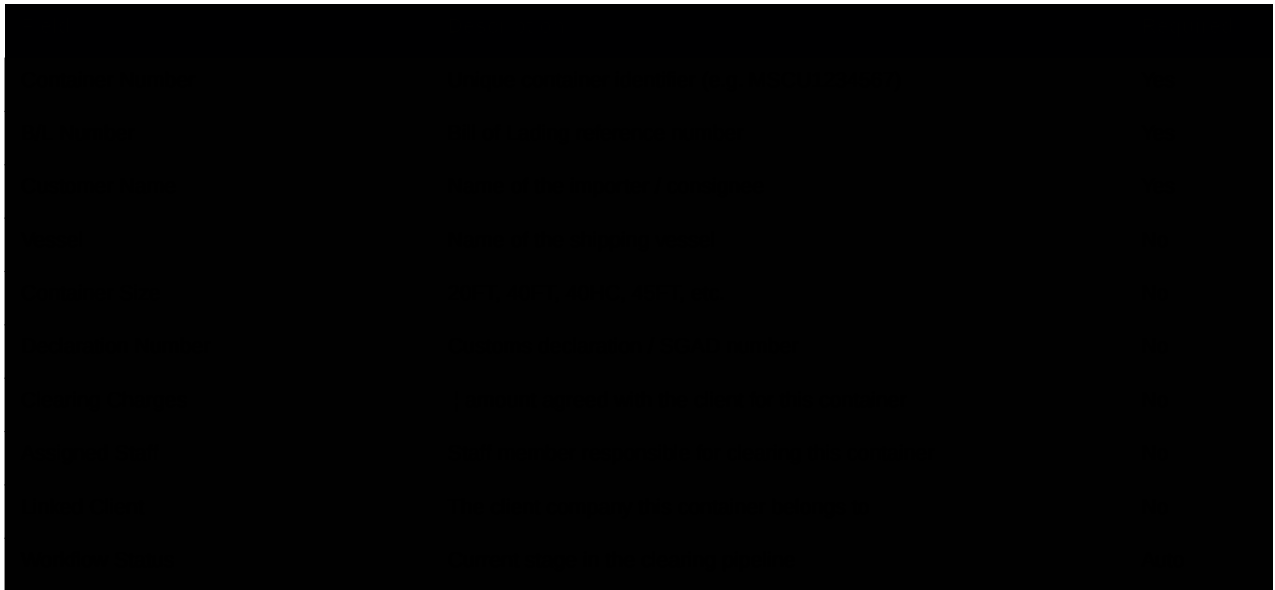
[illegible]



Container Management

Containers are the central entity in the system. Every cost, document, task, approval, timeline entry, and invoice is ultimately linked to a container. Each container represents a physical shipping container going through the bonded terminal clearing process.

Container Record — Key Fields



Workflow Stages

Each container moves through the following pipeline stages:

Registered:	Container has been added to the system; initial intake
Documentation:	Bill of lading, packing list, and customs documents are being processed
Customs Examination:	Customs officer examination is underway
Duty Assessment:	Duty amounts are being assessed by customs
Duty Payment:	Duties are paid and awaiting release
Terminal Processing:	Container is at the terminal being processed for release
Awaiting Delivery:	Container cleared; awaiting transport arrangement
In Transit:	Container is on its way to the final destination
Completed:	Container successfully delivered and process closed
Closed:	File closed (admin action)

Container Detail Tabs

Each container detail page has the following tabs:

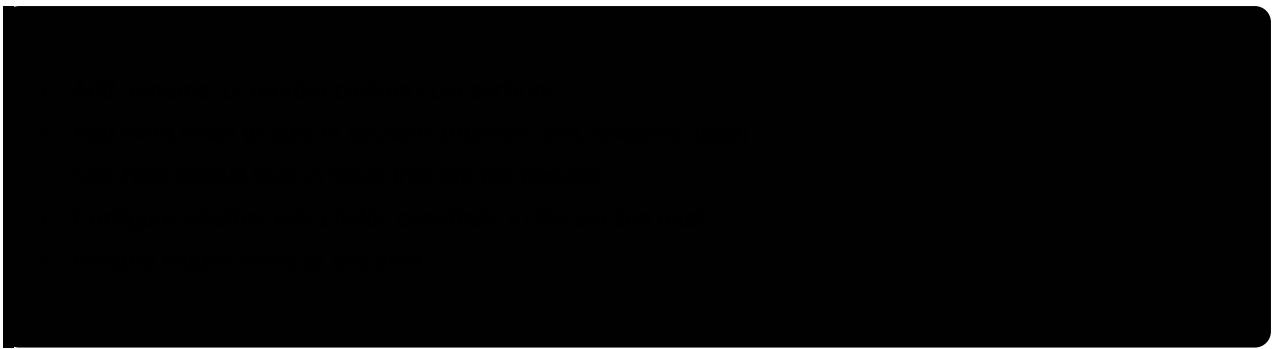
[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]





Cost Recording — Five Built-in Sections

Every container has five built-in cost sections. Each section captures a specific category of clearing charges. All values are in Nigerian Naira (₦). Admins can add extra fields to any section, and can hide default fields that are not applicable.

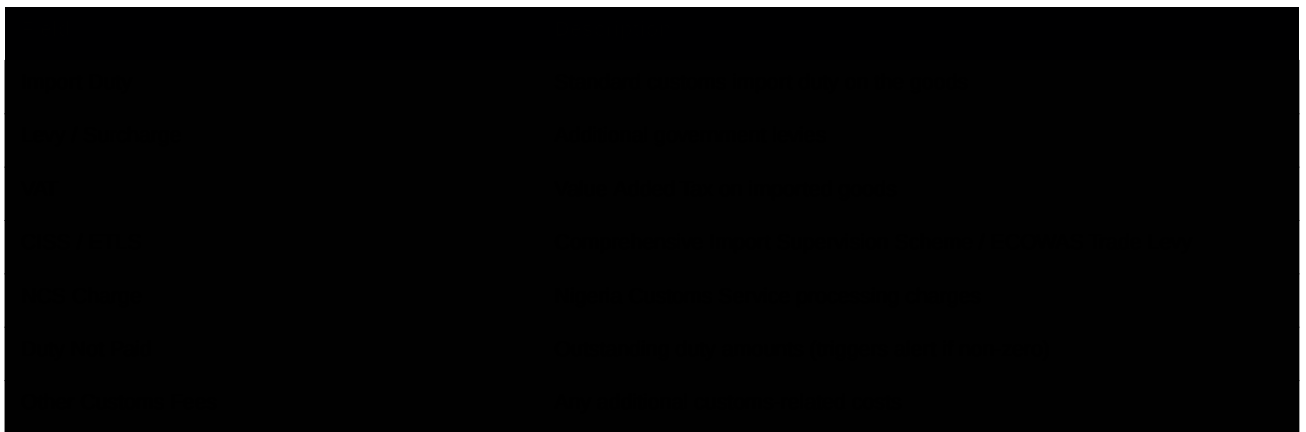
1 — Shipping Charges

Costs related to the shipment itself, port dues, and miscellaneous freight charges.



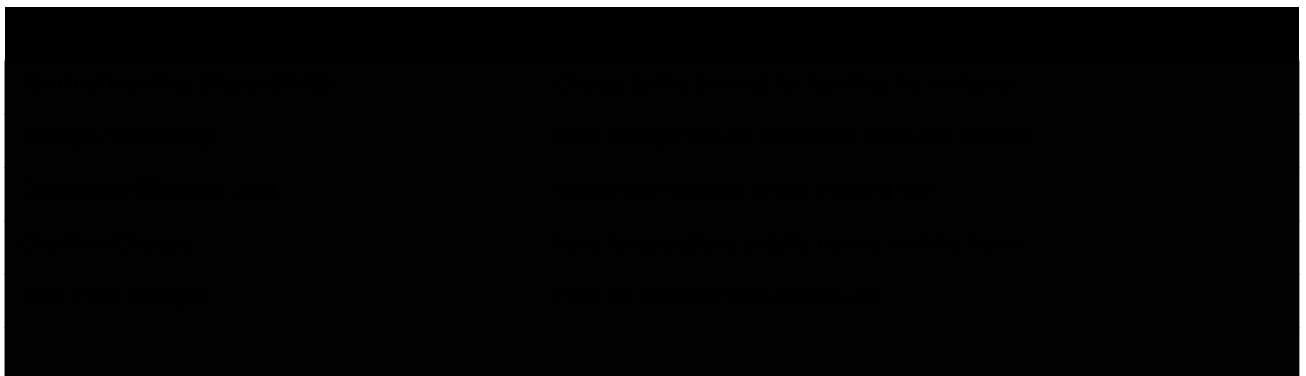
2 — Customs Charges

All government duties, taxes, and customs-related costs.



3 — Terminal Charges

Storage, demurrage, handling, and terminal operator fees.



Shunting Charges

Charges for moving the container within the terminal

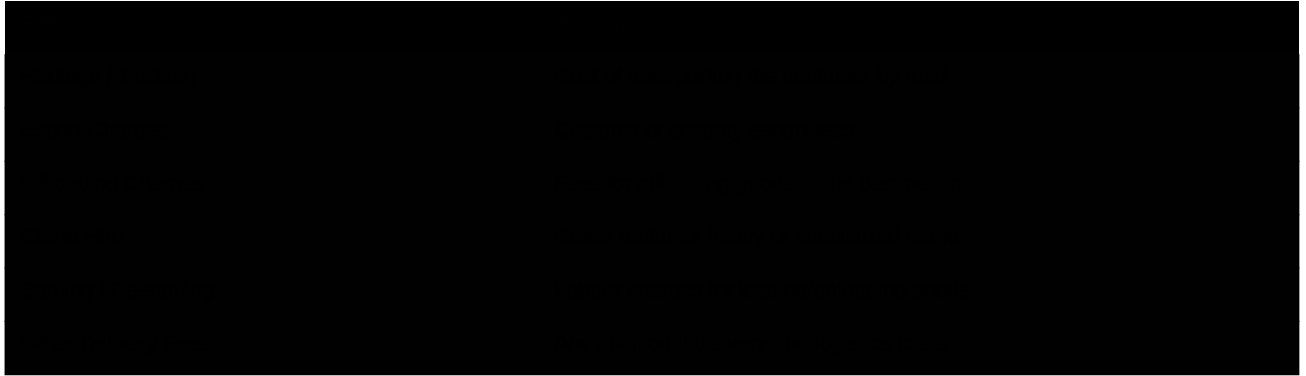


Other Terminal Fees

Any additional terminal-related charges

4 — Delivery Charges

Transportation and haulage costs from the terminal to the final destination.



5 — Operations Charges

Internal and external operational costs incurred during the clearing process.





Custom Sections & Extra Fields

Administrators can fully customize the cost structure of the system. Beyond the five built-in sections, custom sections can be created, and extra fields can be added to any built-in section.

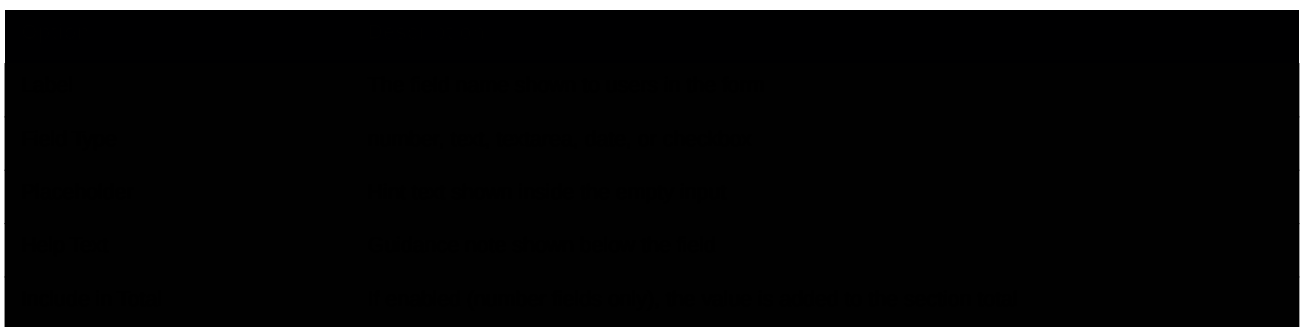
Custom Cost Sections

- Create an unlimited number of additional cost sections (e.g. "Insurance", "Government Agency Fees")
- Name and rename sections at any time
- Add any number of fields to each custom section with the following types:
 - Number — numeric monetary value (\$), optionally included in the section total
 - Text — single-line free-text entry
 - Textarea — multi-line notes or descriptions
 - Date — calendar date picker
 - Checkbox — yes/no toggle field
- Fields can be given a label, placeholder text, and help/guidance text
- Reorder sections using drag-and-drop
- Delete custom sections (with confirmation)

Extra Fields on Built-in Sections

- Any of the five built-in sections can have extra fields added by an admin
- Fields are added via the "Edit Cost Sections" dialog !' expand a section !' "Add field to this section"
- Added fields appear in the container charge form below the standard fields for that section
- Fields can be configured to include their value in the section cost total
- Added fields can be edited or deleted at any time
- Default built-in fields can be soft-hidden (removed from view without deleting data)
- Hidden built-in fields can be restored at any time via the eye/restore icon

Field Configuration Options





Client Management

Clients are the companies and individuals on whose behalf you clear containers. The system tracks their contact information, their containers, their financial position (wallet & invoices), and their payment history.

Client Profile Fields



Adding Clients

- One at a time: Click "Add New Client" and fill the form
- Bulk upload: Download the Excel template, fill it with client data, and upload
- Template columns: CLIENT NAME, CONTACT PERSON, EMAIL, PHONE, ADDRESS, NOTES
- Duplicate detection prevents the same company from being added twice

Client Detail Page

- View all containers linked to this client with filtering and search
- Filter containers by status, date range, or keyword
- View the client wallet (deposits, invoiced amount, and balance)
- View accounts receivable (invoices, payments, outstanding)
- Record deposits into the client wallet
- Create invoices for containers linked to this client
- Download a printable client statement

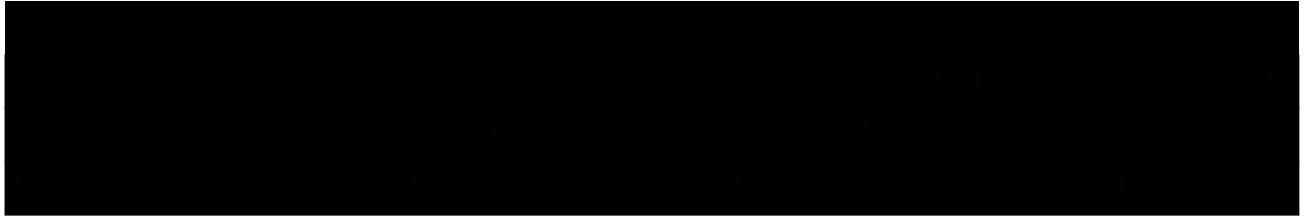
Linking Containers to Clients

- When uploading containers in client mode, all containers in the batch are automatically linked
- Existing containers can be linked to a client from the client detail page
- A container can only be linked to one client at a time
- The customer name on the container is auto-filled from the client name when uploading in client mode

Client Wallet & Financial Tracking

The Client Wallet is a retainer-style financial ledger for each client. Clients deposit funds in advance, and as invoices are issued, those amounts are deducted from the wallet balance. This gives a real-time view of each client's financial position.

Wallet Summary — Three Key Figures



Note: "Total Invoiced" is sourced from the invoices issued to the client — not from raw clearing costs. This ensures the wallet balance is always in sync with the Accounts Receivable section.

Recording Deposits

- Click "Record Deposit" in the wallet card
- Enter the amount, payment method (Cash / Bank Transfer / Cheque), reference number, and notes
- Deposits appear in the deposit history table with date, amount, method, and reference
- Individual deposits can be removed by admins (with confirmation)

Wallet Reset (Admin Only)

Admins can reset the wallet to start fresh from a specific point in time. This is useful at the beginning of a new financial period or when a client relationship is being restructured.

- Click the "Reset Wallet" button (visible to admins only) in the wallet card header
- A security dialog opens requiring TWO confirmations:
 - 1. Type the word RESET exactly in the confirmation field
 - 2. Enter your admin login password (verified server-side)
- The reset sets a new starting point at the current date and time
- All deposits and invoices before the reset date are excluded from the balance calculation
- Historical records are preserved — they are NOT deleted
- The date and time of the last reset is shown in the wallet card
- After reset, the balance starts at \$0.00 until new deposits are recorded

Overdraft Detection

- If invoiced amounts exceed total deposits, the wallet shows an "Overdrawn" state in red
- The balance is shown as a negative amount with an "Outstanding liability" warning message
- This alerts the team to collect funds from the client

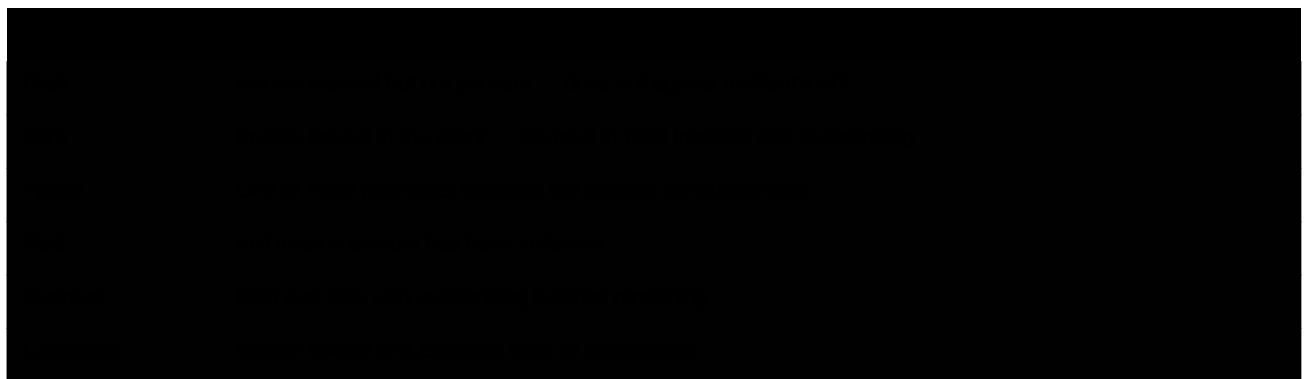
Invoicing & Accounts Receivable

The system has a full-featured invoicing module. Invoices are issued to clients for clearing services, can be sent and tracked, and payments are recorded against them. The Accounts Receivable (AR) module gives a consolidated view of all outstanding amounts across all clients.

Creating an Invoice

- Navigate to Invoices !' "New Invoice"
- Select the client and optionally link to a specific container
- Add line items: each line has a description and | amount
- The system auto-populates a suggested amount based on the container's clearing charges
- Add VAT amount separately if applicable
- Set a due date and add internal notes
- Invoice number is auto-generated (e.g. INV-0042)
- Save as Draft to review before sending, or set to Sent immediately

Invoice Status Lifecycle



Recording Payments

- Open the invoice !' click "Record Payment"
- Enter payment amount, method (Cash / Bank Transfer / Cheque), reference, and date
- Multiple partial payments can be recorded against a single invoice
- Invoice status automatically updates to Partial or Paid based on collected amount
- Payment history is shown on the invoice detail page

Invoice Detail Page

- Full invoice preview showing client info, line items, VAT, and totals
- Payment history with dates, methods, and references
- Edit invoice line items, notes, and due date
- Mark as sent, cancel, or re-open the invoice
- Print the invoice as a formatted PDF (printable view)

Accounts Receivable Module

The AR page (accessible from the sidebar) gives a real-time view of all outstanding invoices across every client — a single screen to understand overall collection position.

- Total Invoiced — all sent/active invoices across all clients

Total Collected — all payments recorded across all invoices

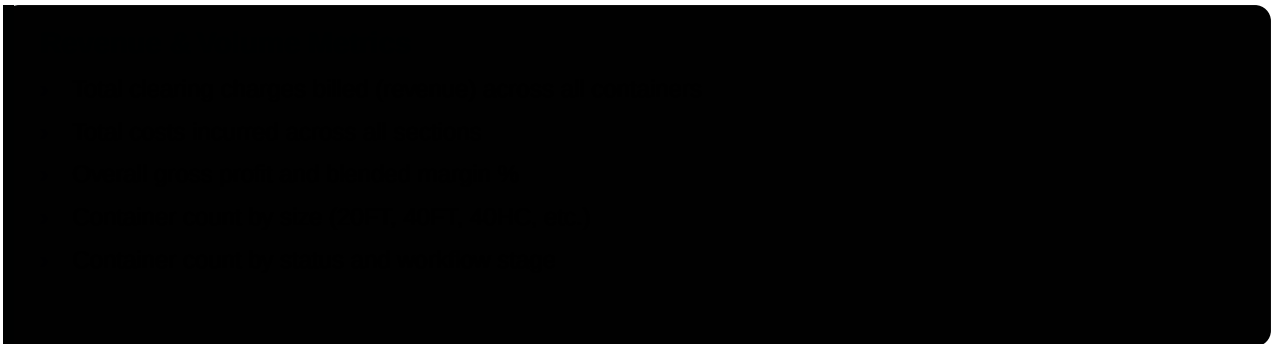
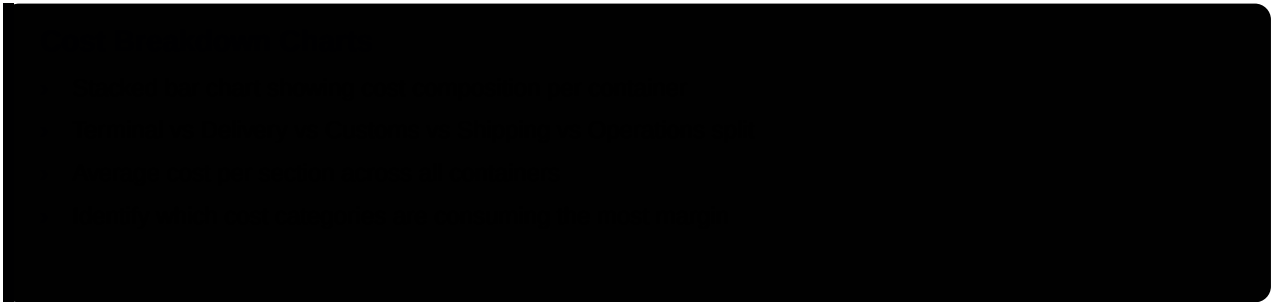
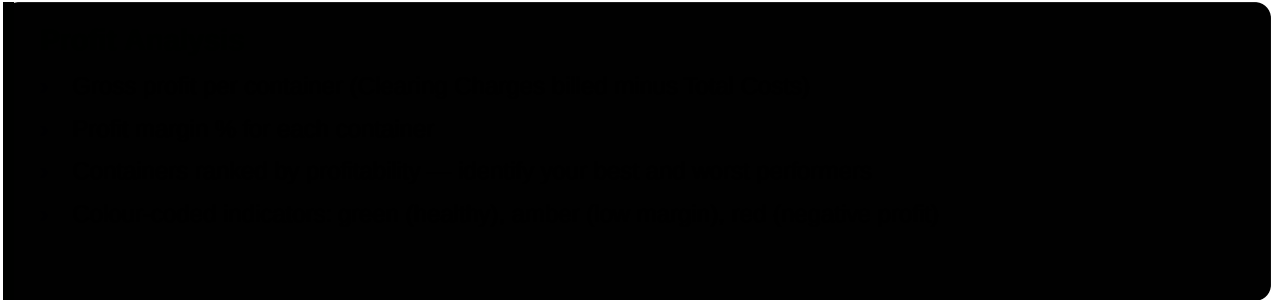
- Total Outstanding — amount not yet collected
- Filter by client, status, or date range
- Click through to any invoice for details or to record a payment
- On each client's detail page, their AR section shows their individual invoiced / collected / outstanding figures



Analytics & Reports

Analytics Dashboard

The Analytics page provides a financial performance overview across all containers in the system. All charts and numbers update in real time from the live database.



Reports Module

Printable / exportable reports for accounting, compliance, and client communication:

Client Statement

- A full financial statement for a specific client over a selected date range
- Shows all containers cleared, costs, invoices issued, payments collected, and outstanding balance
- Printable in a professional format suitable for sending to the client

VAT Summary Report

- Summarises all VAT amounts across invoices for a given period
- Grouped by month for easy reconciliation with FIRS returns
- Shows VAT charged and VAT collected

Invoice Aging Report

Lists all outstanding invoices grouped by how overdue they are

- Aging buckets: Current (not yet due), 1-30 days, 31-60 days, 61-90 days, 90+ days
- Helps prioritise collection efforts

Delivery Report

- A log of all containers that have reached the Completed or Delivered stage
- Includes container number, client, delivery date, and total clearing cost
- Useful for reconciliation and operations review

Filtering & Date Ranges

- All reports support date-from and date-to filtering
- Client filter for client-specific reports
- Status filter for container-specific views
- All report pages include a "Print" button that opens the browser's native print dialog



Pipeline / Kanban Board

The Pipeline page presents all active containers as cards on a Kanban-style board, organised by workflow stage. This gives a visual, at-a-glance overview of where every container is in the clearing process.

Board Columns

Each column on the board corresponds to a workflow stage:

- Registered !' Documentation !' Customs Examination !' Duty Assessment !' Duty Payment
- !' Terminal Processing !' Awaiting Delivery !' In Transit !' Completed !' Closed

Container Cards

- Each card shows the container number, customer name, vessel, size, and assigned staff
- Colour-coded status badge on each card
- Click a card to navigate directly to the container detail page
- Cards showing a red/amber indicator have financial alerts active

Moving Containers

- Drag a container card from one column to another to update its status
- The status change is saved immediately to the database
- A timeline entry is automatically created recording the stage transition with timestamp and user

Filtering the Board

- Filter by assigned staff member — see only your containers or a specific person's pipeline
- Filter by client — see all containers for a specific company
- Search by container number or customer name
- Toggle to show/hide completed and closed containers



Notifications & Alerts

The system continuously monitors all containers and generates alerts when financial or operational thresholds are breached. Alerts are shown in the Notifications page and via a badge on the sidebar icon.

Alert Types



Notification Features

- Notifications page shows all active alerts in a filterable list
- Filter by: All / Unread / Read (toggle buttons)
- Filter by alert type (dropdown)
- Each notification links directly to the affected container
- Unread notifications show a blue dot indicator
- "Mark all read" button clears all unread indicators in one click
- Read/unread state is per-user and persists across sessions
- Sidebar bell icon shows a red badge with the unread count
- Notifications auto-refresh every 60 seconds
- Manual "Refresh" button available for immediate update

Aging Thresholds (Configurable)

- All delay thresholds are configured in Settings !' Aging Thresholds
- Inactivity days: days since last recorded activity before "No Activity" alert fires
- Days 1 / 2 / 3: number of days a container can be in clearing before Warning / High / Critical alerts



Approvals Workflow

The approvals system provides an internal control mechanism for cost sections. Before a section's costs are considered final, they must go through a review and approval process.

How Approvals Work

- Each of the five built-in cost sections (and custom sections) can be submitted for approval separately
- Any user with the "Can Approve" permission (or any admin) can approve or reject
- The user who submitted the section cannot approve their own submission

Approval States



Approval Actions

- Submit for Approval — locks the section and creates a pending approval request
- Approve — marks the section as approved; costs are finalized
- Reject — marks the section as rejected; approver must provide a rejection reason
- Rejected sections can be re-edited and resubmitted for approval
- Full approval history (who submitted, who approved/rejected, when, and comments) is preserved

Approvals Page (Global View)

- The Approvals page in the sidebar shows all pending approval requests across all containers
- Filter by container, section, or submitter
- Approve or reject directly from the list without navigating to each container



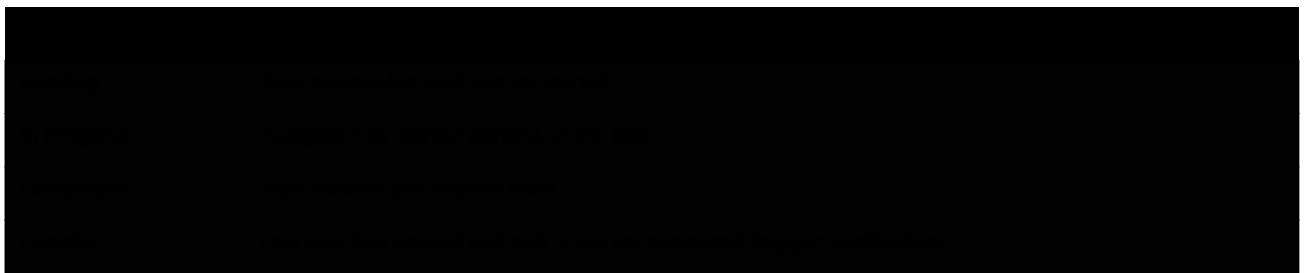
Task Management

Tasks can be created at the container level and assigned to staff members. The "My Tasks" page gives each staff member a personal view of everything assigned to them.

Creating Tasks

- Open any container !' Tasks tab !' "Add Task"
- Enter the task title, description, assignee, and due date
- Tasks can be assigned to any staff member in the system
- Priority levels: Low, Medium, High, Urgent

Task Status



My Tasks Page

- Personal view showing only tasks assigned to the currently logged-in user
- Filter by status: All / Pending / In Progress / Completed / Overdue
- Filter by due date range
- Each task card links to its parent container
- Mark tasks as In Progress or Completed directly from the My Tasks page
- Overdue tasks are highlighted in red

Container Upload — Bulk & Single

Upload Modes

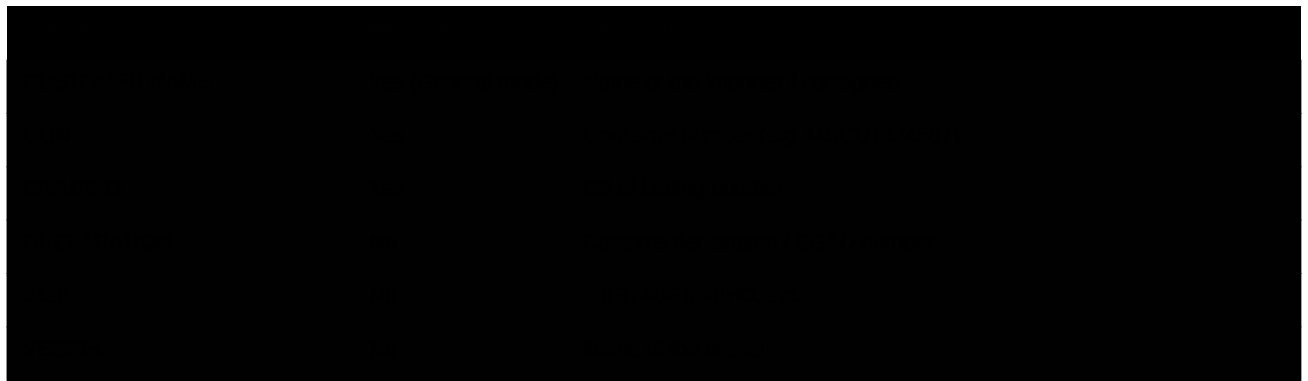
The Upload page supports two modes:

- General Mode — containers are uploaded without being linked to a specific client. Customer names come from the spreadsheet itself.
- Client Mode — select a client first. All containers in the upload are automatically linked to that client. Customer name is taken from the client record.

Supported File Formats

- Excel (.xlsx, .xls)
- CSV (.csv)

Template Columns



Duplicate Detection

- Before importing, the system checks every container number and B/L number against existing records
- Each row is labelled: New (green), Exists in system (amber), or Duplicate in file (orange)
- Rows with conflicts can be selectively excluded from the import by clicking the toggle on that row
- Upload proceeds with only the rows you choose to include

Upload Process

- Drag and drop your file onto the upload zone, or click to browse
- The file is parsed immediately — any formatting errors are listed
- A preview table shows all rows with their status (new / duplicate)
- Review conflicts and decide which rows to include
- Click Upload to import the approved rows
- A success summary shows how many containers were created

Single Container Entry

- Click "New Container" from the Containers list page
- Fill in the container number, B/L, customer name, vessel, size, declaration, clearing charges, and assigned staff
- Save to create the container and navigate to its detail page
- All five cost sections are immediately available for data entry



Public Container Tracking

The system includes a publicly accessible tracking page that allows clients to check the status of their container without needing a login or account.

How It Works

- Share the tracking URL with your client: [your-domain]/tracking/[CONTAINER-NUMBER]
- The client enters or is given the container number directly in the URL
- No authentication is required to access the tracking page
- The page shows the current workflow stage and a summary of the container status

Information Displayed

- Container number and B/L number
- Customer name and vessel
- Current workflow status with a visual stage indicator
- Container size
- Date the container was registered in the system

What Is NOT Shown

- No cost figures are shown on the tracking page (financial data is internal only)
- No staff names or internal notes
- No documents or task details
- The tracking page is read-only — clients cannot make any changes

Use Case

This feature is ideal for sharing real-time status updates with importing clients who want to know where their goods are in the clearing process — without giving them any access to your internal financial data or system.



Settings & Configuration

The Settings page (admin only) allows configuration of key system behaviours.

Aging & Delay Thresholds

Configure when the notification system fires delay-related alerts:

- Inactivity Days — number of days without any recorded activity before the "No Activity" alert fires
- Delay Threshold 1 (Warning) — days in clearing before a "Clearing Delay" warning fires
- Delay Threshold 2 (High) — days before a "Long Delay" alert fires
- Delay Threshold 3 (Critical) — days before a "Critical Delay" alert fires

Default values: Inactivity = 7 days; Delays = 30, 60, 90 days

Cost Section Templates

Manage the global structure of the container cost form:

- Create new custom sections visible on every container's Charges tab
- Rename any custom section
- Add, edit, or delete fields within custom sections
- Add extra fields to any of the five built-in sections
- Hide default built-in fields that are not used in your operations
- Restore previously hidden built-in fields
- Reorder custom sections

User Management

Accessible from the Users page (linked from Settings for admins):

- View all active staff accounts
- Create new staff accounts (name, email, password, role)
- Edit staff permissions: Can Upload, Can Approve
- Reset staff passwords
- Deactivate or delete staff accounts



WhatsApp Notifications

The system integrates with Twilio's WhatsApp API to send automated notifications to designated phone numbers when significant events occur.

Configuration

- Requires Twilio account credentials: Account SID, Auth Token, and WhatsApp From number
- Credentials are stored as secure environment secrets — never exposed in the interface
- Once configured, WhatsApp notifications are sent automatically when triggered

Notification Triggers

- New container uploaded / created
- Container status changes to a key workflow stage (e.g. Awaiting Delivery)
- High-severity financial alerts: negative profit, critical delay
- Invoice created or marked as overdue

Message Format

WhatsApp messages include:

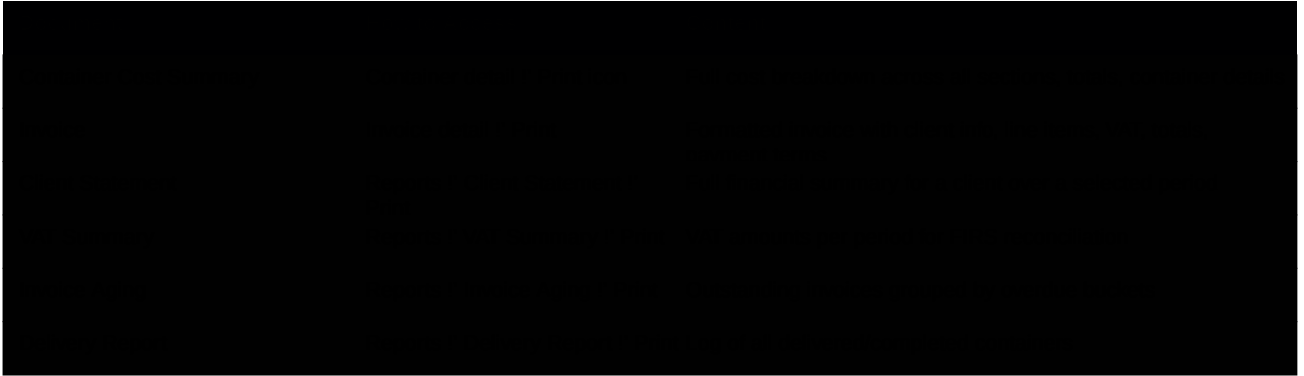
- Alert type and severity
- Container number and customer name
- A short description of the issue or event
- The current timestamp



Print & Export

All major documents in the system have a dedicated print view optimised for PDF export via the browser's native print function (Save as PDF).

Printable Documents



How to Save as PDF

- Click the Print button on any print page
- In the browser print dialog, set Destination to "Save as PDF"
- Select paper size (A4 recommended), margins, and orientation as needed
- Click Save to download the PDF file



Security & Data Integrity

Authentication & Session Security

- All users must log in with email and password before accessing any data
- Passwords are hashed using bcrypt — never stored in plain text
- Sessions are server-managed with HTTP-only cookies — resistant to XSS
- Inactive sessions expire automatically
- Admin password is re-required for destructive operations (e.g. wallet reset)

Role-Based Access Control

- Every API endpoint checks the user's role and permissions before processing
- Admin-only endpoints reject non-admin requests with a 403 Forbidden response
- Staff users can only modify containers assigned to them

Audit Logging

- All significant actions are logged with a timestamp and the user's identity
- Audit log entries are attached to the relevant container
- Actions logged include: cost saves, status changes, document uploads, approval decisions, wallet operations
- The audit log is read-only — entries cannot be edited or deleted

Data Integrity

- Duplicate container numbers are rejected at upload and single-entry time
- All monetary values are stored as high-precision numeric types (15 digits, 2 decimal places)
- Foreign key constraints ensure referential integrity across all tables
- Soft-delete patterns are used where appropriate — data is hidden, not permanently destroyed
- The wallet reset stores a timestamp rather than deleting historical deposit records

Sensitive Data Handling

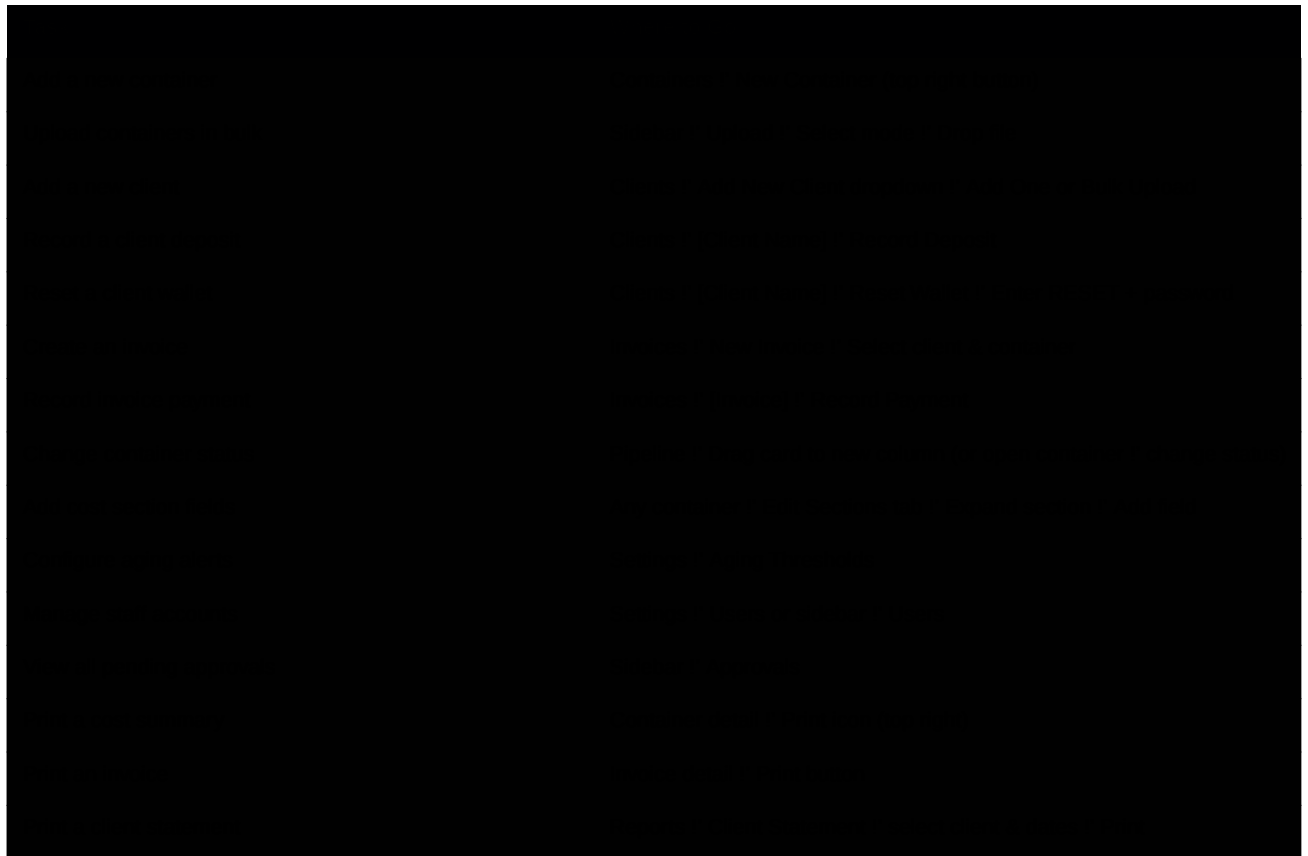
- API credentials (Twilio) are stored as environment secrets — not accessible via the UI
- The public tracking page exposes only status information — no financial or personal data
- Client financial data (invoices, wallet balance) is only visible to logged-in users

Quick Reference — Key Shortcuts & Tips

Navigation

- Use the sidebar to navigate between all main sections
- Breadcrumb links appear at the top of detail pages for quick back-navigation
- The sidebar notification badge shows your unread alert count at all times

Common Admin Tasks



Keyboard Tips

- Press Escape to close any open dialog
- Press Enter in password fields to submit (e.g. wallet reset dialog)
- Use Tab to move between form fields for faster data entry

